

CAREER PROGRESSION AND PROMOTION GAP ANALYSIS FOR RETENTION OPTIMIZATION

A Data Science Research / Technical Project Paper

Domain: Human Resources Analytics

Dataset: Palo Alto Networks.csv (project-provided dataset)

Author: Manani Pratik

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Streamlit

Abstract

This project analyzes employee career progression and promotion-gap patterns to support proactive retention decision-making. The project uses a workforce dataset containing employee demographic, job, compensation, satisfaction, training, tenure, promotion and attrition-related attributes. The analytical workflow covers data validation and cleaning, exploratory data analysis, feature engineering, unsupervised career-path clustering, promotion-gap risk scoring and retention opportunity identification. Derived indicators include promotion gap ratio, role stagnation ratio, training intensity and managerial stability. The final analytical outputs are intended to be consumed through an interactive Streamlit dashboard.

The current repository contains 1,470 employee records. The latest Day 6 outputs classify 1,001 employees as low promotion-gap risk, 459 as medium risk and 10 as high risk. A separate promotion-gap analysis identifies 260 employees in the long-promotion-gap group, representing 17.69% of the dataset. However, the current Day 6 output reports an observed attrition rate of 0% and zero medium/high retention opportunity employees. These values require validation before they are presented as business findings, because they may indicate an upstream attrition/retention-label issue rather than a genuine workforce pattern.

Keywords: HR Analytics, Career Progression, Promotion Gap, Employee Retention, Career Clustering, Risk Scoring, Workforce Analytics, Streamlit

1. Introduction

Employee retention is influenced not only by compensation or immediate job satisfaction but also by career progression, role mobility, training and managerial continuity. Traditional attrition analysis answers an important question—who may leave—but does not always explain the career structure behind employee disengagement. This project therefore focuses on career trajectory intelligence as a complementary retention approach.

The project is designed to identify employees and workforce segments showing promotion delays, long periods in the same role, limited development signals or other career-stagnation indicators. Instead of treating a risk score as a prediction of employee behavior, the system is designed as a transparent decision-support framework for HR review.

1.1 Problem Statement

The project addresses the lack of a data-driven view of career progression patterns, identification of promotion stagnation, and early identification of retention opportunities before employee disengagement becomes visible through attrition.

1.2 Objectives

1. Analyze employee workforce and career-progression patterns.
2. Quantify promotion gaps and role-stagnation signals.
3. Engineer interpretable career-development indicators.
4. Segment employees into career-path clusters using unsupervised learning.
5. Develop a promotion-gap risk indicator with Low/Medium/High categories.
6. Identify retention opportunities for proactive HR intervention.
7. Deliver results through an interactive Streamlit dashboard.

2. Dataset and Data Understanding

The project uses the file Palo Alto Networks.csv supplied for the internship project. The repository currently stores the raw dataset along with cleaned, feature-engineered, clustering and final risk-analysis outputs. The public repository currently contains the six analysis notebooks used across the project workflow.

Field	Business Meaning
Age	Employee age in years
Attrition	Whether the employee left the organization
Department	Department in which the employee works
JobRole	Specific job role/designation
JobLevel	Seniority level
YearsAtCompany	Years employed by the organization
YearsInCurrentRole	Years in the current role
YearsSinceLastPromotion	Years elapsed since the last promotion
YearsWithCurrManager	Years working with the current manager
TrainingTimesLastYear	Number of training programs attended in the last year
PerformanceRating	Employee performance evaluation score
JobSatisfaction	Satisfaction with the job role
WorkLifeBalance	Work-life balance rating

The project repository currently contains the raw dataset and downstream data products including feature_engineered_palo_alto.csv and hr_career_risk_analysis.csv, providing a reproducible transition from raw data to analytical outputs.

3. Data Science Methodology

Stage 1 – Business Understanding: Define the business problem around career progression, promotion gaps, stagnation and proactive retention.

Stage 2 – Data Validation and Cleaning: Inspect schema, missing values, duplicates, data types, categorical consistency and potential outliers before analysis.

Stage 3 – Exploratory Data Analysis: Analyze workforce composition, career tenure, promotion timing, role duration, satisfaction and other HR dimensions.

Stage 4 – Feature Engineering: Create career-oriented indicators that make raw tenure and promotion variables more comparable across employees.

Stage 5 – Career Path Clustering: Use unsupervised learning to identify groups of employees with similar career trajectories.

Stage 6 – Risk and Retention Analysis: Combine career-stagnation signals into interpretable risk categories and identify potential retention opportunities.

Stage 7 – Dashboard: Present the analytical results through an interactive Streamlit interface with filtering, KPIs, career progression views, cluster views and risk/retention outputs.

Methodology Flow

Step	Output
Business Understanding	Problem definition and KPIs
Cleaning	Validated analytical dataset
EDA	Workforce and career insights
Feature Engineering	Career progression indicators
Clustering	Career-path segments
Risk Scoring	Promotion-gap risk levels
Retention Analysis	Retention opportunity indicators
Dashboard	Interactive HR decision-support tool

4. Feature Engineering

The project requirements define several derived measures intended to capture career progression more directly than the raw fields.

Feature	Formula / Definition	Purpose
Promotion Gap Ratio	$\text{YearsSinceLastPromotion} / \text{YearsAtCompany}$	yRelative promotion delay
Role Stagnation Ratio	$\text{YearsInCurrentRole} / \text{YearsAtCompany}$	Relative time in current role
Training Intensity	$\text{TrainingTimesLastYear} / \text{YearsAtCompany}$	Training participation relative to tenure
Manager Stability	Derived from manager-tenure information	Context for managerial continuity

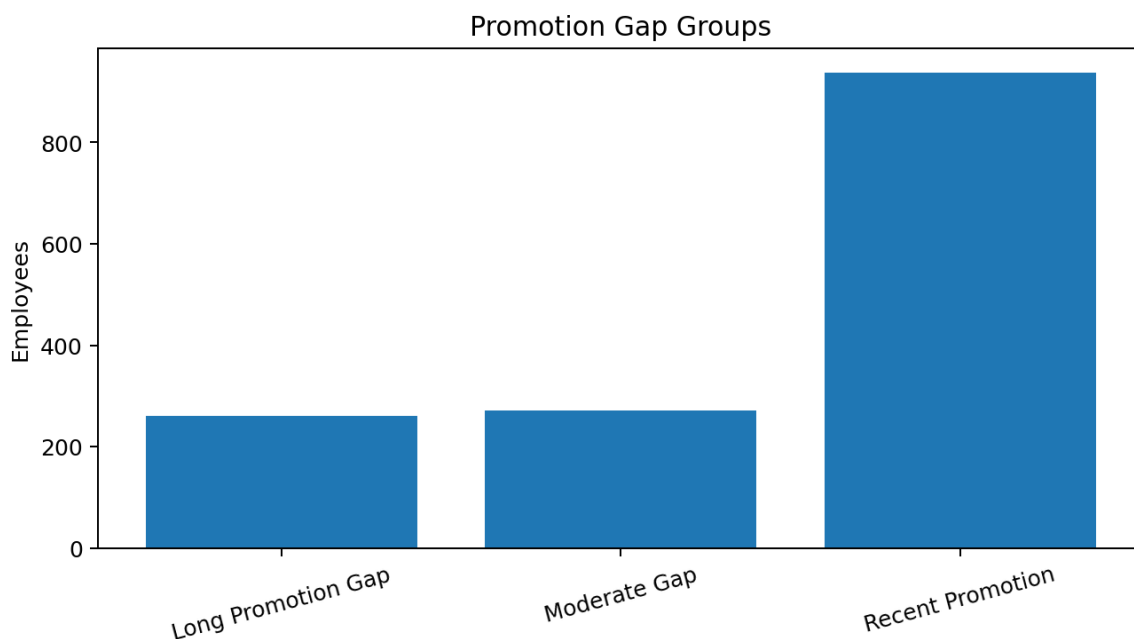
These features convert raw career-history variables into ratios or indicators that can be compared across employees with different levels of organizational tenure.

4.1 Why Feature Engineering Matters

An employee with eight years since promotion and ten years at the company has a different relative career profile from an employee with eight years since promotion and twenty years at the company. Ratio-based features therefore provide additional context beyond absolute years.

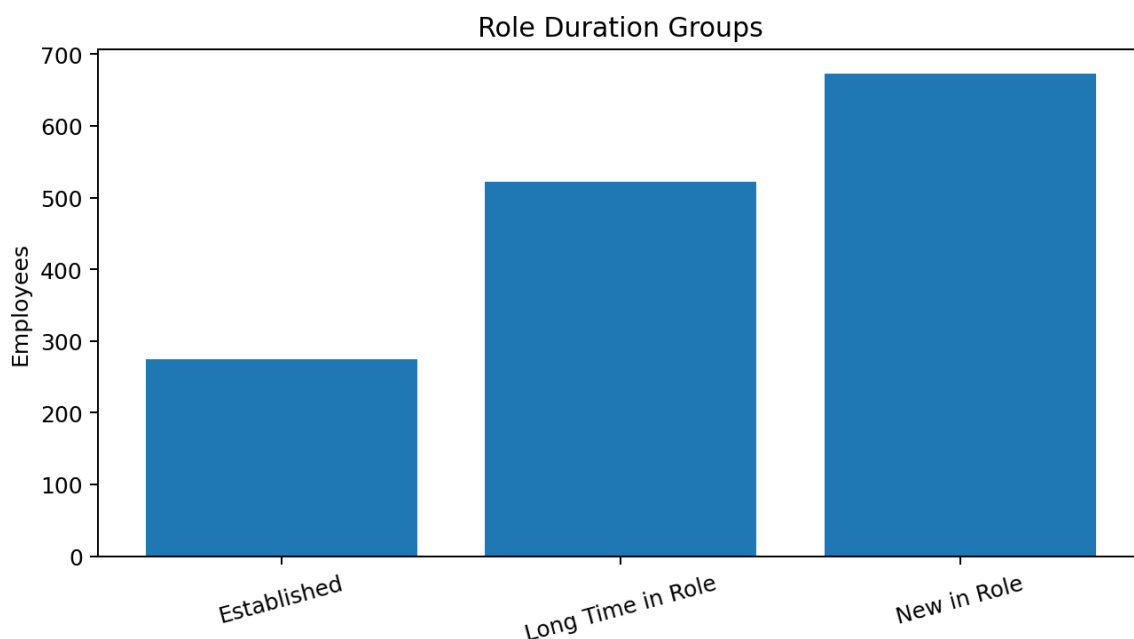
5. Exploratory Analysis and Current Findings

5.1 Promotion Gap Distribution



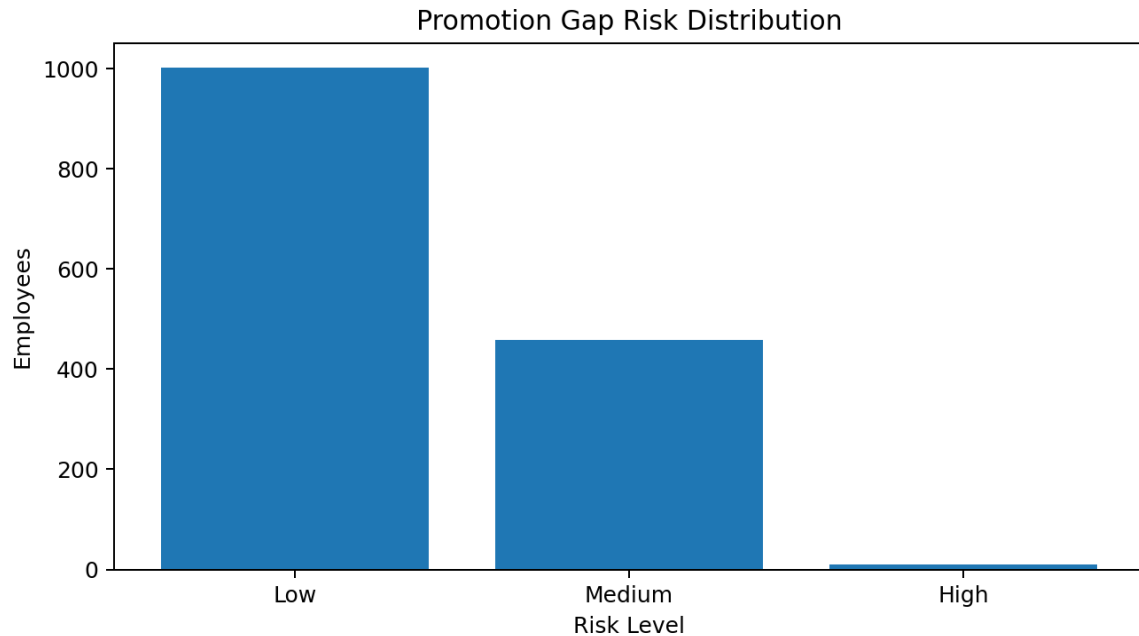
The current output classifies 938 employees (63.81%) as having a recent promotion, 272 (18.50%) as moderate-gap employees and 260 (17.69%) as long-promotion-gap employees. The long-gap group has an average of approximately 8.23 years since promotion. These results provide a useful career-progression screening signal.

5.2 Role Duration



The role-duration output shows 522 employees (35.51%) in the 'Long Time in Role' group, with an average of about 8.46 years in the current role. A further 673 employees (45.78%) are classified as new in role, while 275 (18.71%) are established. This indicates substantial variation in career-stage positioning.

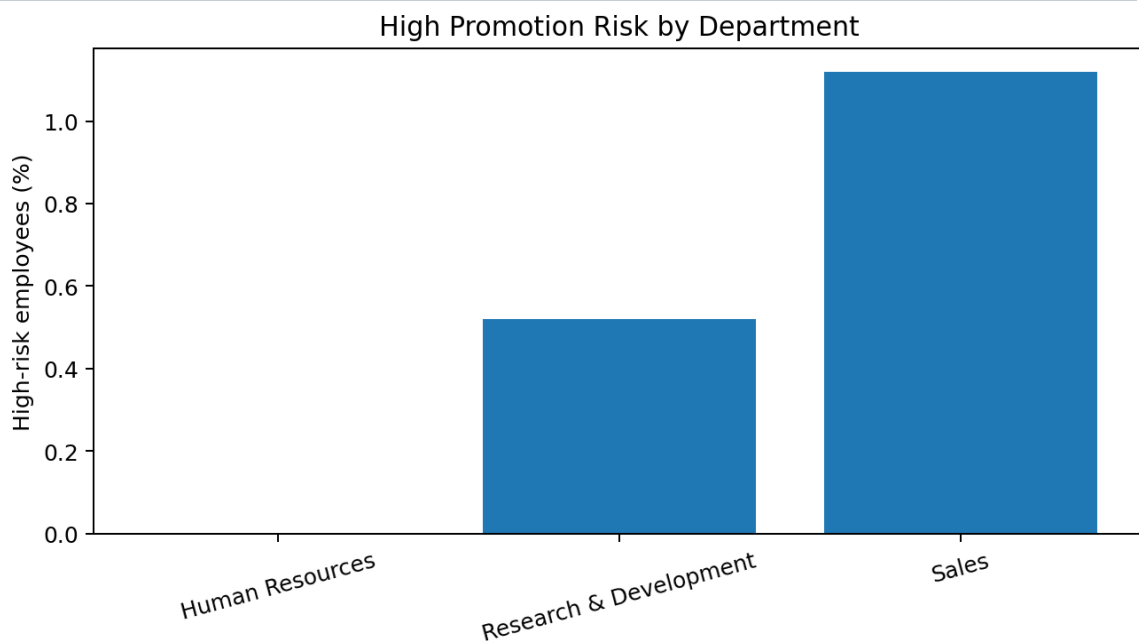
6. Promotion Gap Risk Analysis



The current Day 6 risk output contains 1,001 low-risk employees (68.10%), 459 medium-risk employees (31.22%) and 10 high-risk employees (0.68%). The high-risk category is intentionally small under the current scoring framework and should therefore be interpreted as a targeted review group rather than a prediction that these employees will leave.

6.1 Department-Level Risk

Department	Employees	High Risk	High Risk %
Human Resources	63	0	0.00%
Research & Development	961	5	0.52%
Sales	446	5	1.12%



Sales has the highest high-risk percentage among the three departments at 1.12%, followed by Research & Development at 0.52%. Human Resources has no employees in the high-risk category in

the current output.

7. Job-Role Risk Analysis

Job Role	Employees	High Risk	High Risk %
Healthcare Representative	131	1	0.76
Human Resources	52	0	0.0
Laboratory Technician	259	0	0.0
Manager	102	1	0.98
Manufacturing Director	145	2	1.38
Research Director	80	1	1.25
Research Scientist	292	1	0.34
Sales Executive	326	4	1.23
Sales Representative	83	0	0.0

Among the current role-level outputs, Manufacturing Director has the highest percentage of high-risk employees (1.38%), followed by Research Director (1.25%) and Sales Executive (1.23%). Sales Executive also has the largest absolute number of high-risk employees in the role table, with four employees.

These figures should be used to prioritize investigation and career-review conversations, not to make individual employment decisions automatically.

8. Retention Opportunity Analysis

Opportunity Level	Employees	Percentage
Low	1,470	100.0%
Medium	0	0.0%
High	0	0.0%

The current retention-opportunity output classifies all 1,470 employees as Low opportunity. This result is not sufficiently informative for a final business conclusion and should be treated as a validation issue in the current implementation. The project should revisit the opportunity-score thresholds and input logic before submission.

8.1 Attrition Validation — Critical Data Quality Note

The current Day 6 report records an observed attrition rate of 0.0% and zero attrition observations across all three promotion-risk groups. The same report records a chi-square p-value of 0.013009 and labels the result as evidence of association.

9. Streamlit Dashboard

The project includes a Streamlit dashboard designed to make the analysis accessible to HR stakeholders. The dashboard is organized into five major modules.

Module	Purpose
Executive Overview	High-level workforce, attrition and promotion-risk KPIs
Workforce Analysis	Department, role, level, age and career-stage views
Career Progression	Promotion gap, role duration and progression indicators
Career Clusters	Cluster distribution and career-path profiles
Risk & Retention	Risk levels, retention opportunity and employee-level review

The dashboard also provides sidebar filtering and an employee-level CSV download for filtered results. The implementation is designed as decision support rather than automated employee classification.

9.1 Recommended Dashboard Validation

Before deployment, dashboard KPIs should be reconciled with notebook outputs for total employees, promotion risk distribution, department risk, career clusters and retention opportunity. The final dashboard should read the same final analytical dataset used to produce the Day 6 tables.

10. Business Recommendations

Based on the current outputs, the following recommendations are appropriate as analytical actions:

1. Promotion Review: Review employees in the long-promotion-gap group, especially where long promotion gaps coexist with long time in the current role.
2. Career Development: Use training participation and career-stage indicators to identify potential development opportunities.
3. Department Review: Prioritize deeper investigation of Sales because its current high-risk rate is the highest among the departments in the output.
4. Role Review: Examine roles such as Manufacturing Director, Research Director and Sales Executive for structural promotion-path patterns.
5. HR Validation: Validate the attrition and retention-opportunity logic before using the outputs for operational decisions.
6. Human-in-the-loop Decision Making: Use the model outputs as prompts for HR review rather than automatic promotion, retention or employment decisions.

10.1 Limitations

The analysis is based on the project-provided dataset and the variables available in it. The current pipeline does not establish causal relationships between career progression and attrition. The clustering labels are analytical segments rather than official HR categories. Risk and retention scores are decision-support indicators unless they are validated against future outcomes. Most importantly, the current attrition output requires correction/validation before it can support an attrition-related claim.

11. Conclusion

This project demonstrates a career-intelligence approach to HR analytics by moving beyond a simple attrition question toward an examination of promotion timing, role duration, training, managerial context and career-path patterns. The current outputs show a measurable long-promotion-gap population and a small targeted high-risk group that can be investigated through HR career-review processes.

The project also demonstrates an end-to-end data science workflow: business understanding, data preparation, EDA, feature engineering, unsupervised clustering, explainable risk scoring and dashboard delivery. The Streamlit layer provides a practical interface for exploring the analytical outputs.

For the final internship submission, the most important remaining technical task is validation of the attrition and retention-opportunity calculations. Once those outputs are corrected and reconciled with the dashboard, the project can provide a stronger and more defensible basis for proactive career-development and retention analytics.

12. Future Scope

Future work could include longitudinal employee records, actual promotion-event histories, validated future attrition labels, model calibration, fairness analysis, explainability methods and controlled evaluation of HR interventions. A future version could also track changes in career risk over time rather than relying on a single employee snapshot.

13. Project References and Sources

The paper was prepared from the current public project repository and its saved analytical outputs. The repository contains six notebooks covering business understanding, data validation/cleaning, EDA, feature engineering, clustering and risk/retention analysis. It also contains final analytical CSV outputs and the Day 6 risk/retention report.

Repository: github.com/mananipratik/i_project

Primary analytical outputs used in this paper: `executive_summary.csv`, `promotion_gap_summary.csv`, `department_risk_summary.csv`, `jobrole_risk_summary.csv`, `role_duration_summary.csv`, `retention_opportunity_summary.csv`, `risk_vs_attrition.csv` and `day6_risk_retention_report.md`.

Submission note: This document is a project research/technical paper draft based on the repository state available at the time of generation. It should be updated after the Attrition and Retention Opportunity logic is validated.